وثيقة المراجعة والتلخيص الأكاديمي

Course: Advanced Software Engineering (CS603)
Scope source: Dr. Ali Fahim's requirements vocabulary message, 2026-09-30
Status: All 6 Question & Answer packs are authored, cross-reviewed, strict-linted, and have current DirectWrite PDFs (63 pages total, excluding this index). The pack map accounts for Dr. Ali Fahim's 7 vocabulary areas, all 36 textbook exercises, all 25 slide MCQs, and 17 gap-closing synthesis items. Cross-review included targeted source checks and visual/layout corrections; it was not a blind claim-by-claim audit.

Completed Deliverables Summary

Pack	File	Items	Pages	Scope & Source Grounds	Gates
T1: MCQ	T1_MCQ.md / .pdf	35	10	25 slide MCQs (slides 121–133) with independently checked keys + 10 gap-closing synthesis MCQs	Strict Linter ✅ Current PDF ✅
T2: Descriptive	T2_Descriptive.md / .pdf	21	16	18 textbook stems + 3 synthesis items. Natural page flow, discrete / .	Strict Linter ✅ Current PDF ✅
T3: Compare & Analyze	T3_Compare_Analyze.md / .pdf	14	10	11 textbook stems + 3 synthesis analytical items. Natural page flow.	Strict Linter ✅ Current PDF ✅
T4: Diagrams	T4_Diagrams.md / .pdf	5	7	5 visual modeling topics; original vector slides rendered at 4× for fixed diagrams; two task-specific custom illustrations retained.	Strict Linter ✅ Current PDF ✅
T5: Apply & Create	T5_Apply_Create.md / .pdf	9	13	8 textbook applied scenarios + 1 synthetic SRS audit drill. Seven source-native drawing exercises include constructed diagrams.	Strict Linter ✅ Current PDF ✅
V3: Vocabulary	V3_Vocabulary_Coverage.md / .pdf	7	7	All 7 vocabulary areas from Dr. Ali Fahim's 2026-09-30 message.	Strict Linter ✅ Current PDF ✅
Total	**6 Packs**	**91**	**63**	**36/36 textbook exercises and 25/25 slide MCQs mapped; see source and review limits below.**	**Publication and cross-review gates passed**

Print-ready combined pack: Ch03_Complete_Print_Pack.pdf — 63 pages, in pack order; this index is excluded.

Confirmed scope coverage map

Instructor vocabulary	Primary course evidence	Main exercise forms	Destination Pack
Requirements Engineering: crucial process steps	K.K. scan §3.1, printed pp. 40–41; Fig. 3.1 p.41; slides 2–4	Explain, order/diagram, analyze breakdown	T2, T4, V3
Present state of practice	K.K. scan §3.1, slides 4–5	List challenges, explain consequences, propose modest improvements	T2, T3, V3
Functional and non-functional requirements	K.K. scan §3.1.1 and following types section; slides 6–7	Define, distinguish, classify examples	T1, T2, T3, V3
User and system requirements	K.K. slides 8; written exercise 3.36(b)	Compare, classify	T2, T3, V3
Interface specification	K.K. slide 9	List/explain interface forms and examples	T2, V3
Feasibility Study: purpose and focus	K.K. slides 11–15; slide 10 has an IBM statistic without a report date/citation and is not treated as current evidence	Explain purpose/focus; reason through feasibility factors	T2, T3, T5, V3
Requirements Elicitation: interviews, brainstorming, FAST, QFD, Use Case Approach	K.K. slides 16–19, 27–35; source exercises 3.7–3.14	Define/list and compare elicitation techniques; conduct interviews; model use cases	T1, T2, T3, T4, T5, V3
Requirements Elicitation: Use Case Approach	K.K. slides 27–81; slides 28–35 for concepts/guidelines and 36–81 for worked examples; exercises 3.11–3.14	Define, components, guidelines, use-case diagram/narrative	T2, T4, T5, V3
Requirements Analysis: DFD	K.K. slides 82–86; exercises 3.20–3.23	Explain rules/symbols; draw and analyze DFDs	T2, T4, T5, V3
Requirements Analysis: Data Dictionary	K.K. slides 87–88; exercise 3.22(ii); compare B.B. 2010 §3.3	Define structure/content/benefit	T2, T4, V3
Requirements Analysis: ER diagrams	K.K. slides 89–100; exercises 3.15–3.19, 3.24	Cardinality/participation, associative entity, draw ERD	T2, T3, T4, T5, V3
Requirements Analysis: Software Prototyping	K.K. slide 116; exercises 3.24, 3.26–3.28	Explain, evaluate throw-away/reuse claims, advise on prototype use	T2, T3, T5, V3
Documentation: nature of SRS	K.K. slides 101–103 and scan §3.6–3.7; B.B. 2010 §3.6–3.7 as support	Define purpose, “what” boundary, uses/benefits	T2, V3
Characteristics of a good SRS	K.K. scan §3.6.2, printed pp. 73–75; slides 103–110; B.B. 2010 §3.10	List/explain; detect ambiguity, omission, inconsistency	T2, T3, T5, V3
Organization of SRS	K.K. slides 105–110 and §3.6–3.7; exercises 3.30–3.35	Explain sections/issues; create or critique a short example	T2, T3, T5, V3

Instructor vocabulary	Primary course evidence	Main exercise forms	Destination Pack
Validation: Requirements Reviews	K.K. slides 111–115; B.B. 2010 §3.8	Explain checks/review flow and corrective follow-up	T2, T3, V3

Source identity, hierarchy, and limitations

- **Course primary:** K.K. Aggarwal & Yogesh Singh, *Software Engineering*, 3rd edition. The local scan is `02_Raw_Materials/Reference_Textbooks/AliFahim_Software_Engineering.pdf`; its cover/title page is missing, so identity is high-confidence from internal contents/pagination plus the matching companion slide series. Chapter 3 is printed pp. 40–138, scanned spreads 26–75.
- **Companion slides:** K.K. Aggarwal & Yogesh Singh, New Age International Publishers, 2007, Chapter 3, 140 slides. Slides 121–133 contain 25 MCQs; slides 134–140 contain 35 written stems and skip 3.25. The scanned textbook p.138 contains the missing-in-slides Exercise 3.25, so the full textbook exercise set has 36 stems. Slide MCQs 3.23 also repeat a printed label.
- **Supporting book:** B.B. Agarwal, S.P. Tayal & M. Gupta, 2010, Chapter 3 printed pp. 53–84 (PDF pp. 66–97). It differs in some terminology and adds Requirements Management; keep claims source-specific.
- **Answer key:** no official solution key was provided for the 25 K.K. slide MCQs. Every answer key in T1 was independently checked and verified against K.K. text and diagrams.
- **External question banks and SRS samples** are secondary practice/format material; they do not define lecture scope or establish the professor's answer key.
- **Cross-review (2026-10-17):** Three parallel reviews covered T1, T2–T3, and T4–T5/V3, followed by a coordinator check of the changed source locators, PDF freshness, blank pages, build-tag leakage, and the revised DFD. Corrections included source/page anchors, unsupported generalizations, data-dictionary notation, DFD/ERD assumptions, vocabulary attribution, and applied-diagram layout. The wording of Exercise 3.30 points to §3.4.2, but that locator does not match the scanned pages 56–57; the answer uses the verified SRS-characteristics section §3.6.2, pp. 73–75. This was a targeted cross-review, not a blind verification of every individual claim.

Textbook Exercise & Native Slide Source Accounting

Every native item from the textbook (36 exercises) and companion slides (25 MCQs) is accounted for: - **Slide MCQs 1 to 25:** All 25 located in `T1_MCQ.md` (Items 1 to 25). Note: slide labels repeat 3.23; disambiguated as MCQ-24 and MCQ-25. - **Exercises 3.1 to 3.36:** - 3.1 (Descriptive) → T2 Item 01 - 3.2 (Compare/Analyze) → T3 Item 01 - 3.3 (DFD Modeling) → T4 Model 03 - 3.4 (Compare/Analyze) → T3 Item 02 - 3.5 (Descriptive) → T2 Item 02 - 3.6 (Analysis) → T3 Item 03 - 3.7 (Descriptive) → T2 Item 03 - 3.8 (Descriptive) → T2 Item 04 - 3.9 (Compare/Analyze) → T3 Item 04 - 3.10 (Descriptive) → T2 Item 05 - 3.11 (Descriptive) → T2 Item 06 - 3.12 (Use Case Diagram) → T4 Model 02 - 3.13 (Applied Case) → T5 Drill 01 (Library Management) - 3.14 (Applied Case) → T5 Drill 02 (Railway Reservation) - 3.15 (Descriptive) → T2 Item 07 - 3.16 (Compare/Analyze) → T3 Item 05 - 3.17 (Descriptive) → T2 Item 08 - 3.18 (Compare/Analyze) → T3 Item 06 - 3.19 (Applied Case) → T5 Drill 03 (Airline Reservation ERD) & T4 Model 04 - 3.20 (Applied Case) → T5 Drill 04 (Conflict-Free Resource Assignment DFD) - 3.21 (Applied Case) → T5 Drill 05 (University Examination Results DFD) - 3.22 (Descriptive) → T2 Item 09 (DFD & Data Dictionary Technical Notes) - 3.23 (Applied Case) → T5 Drill 06 (Library Book Borrowing &

Reservation DFD) - 3.24 (Applied Case) → T5 Drill 07 (Hotel Reception System ERD) - 3.25 (Descriptive - Book printed p.138) → T2 Item 10 (Throw-Away Prototyping Rationale) - 3.26 (Descriptive) → T2 Item 11 - 3.27 (Descriptive) → T2 Item 12 - 3.28 (Descriptive) → T2 Item 13 - 3.29 (Compare/Analyze) → T3 Item 07 - 3.30 (Descriptive) → T2 Item 14 - 3.31 (Descriptive) → T2 Item 15 - 3.32 (Descriptive) → T2 Item 16 - 3.33a (Compare/Analyze) → T3 Item 08 - 3.33b (Compare/Analyze) → T3 Item 09 - 3.34 (Applied/Construct) → T5 Drill 08 (Inconsistent SRS Construction) - 3.35 (Descriptive) → T2 Item 17 - 3.36a (Descriptive) → T2 Item 18 - 3.36b (Compare/Analyze) → T3 Item 10 - 3.36c (Compare/Analyze) → T3 Item 11

Current gates

- [x] Locate the professor's exact Ch.3 vocabulary in the supplied transcript.
- [x] Identify the three distinct course-related source families and separate their authority.
- [x] Verify the K.K scan's opening and closing printed-page mapping visually.
- [x] Inventory native MCQ and written-exercise counts and numbering anomalies (all 36 exercises + 25 MCQs).
- [x] Map every term to exact printed pages/slides and verify source statements.
- [x] Map and inspect the diagrams (5 custom 2x vector diagrams + 15 companion slide extractions).
- [x] Check external question-bank patterns without treating them as course authority.
- [x] Extend the local vector collection for Chapter 3 (296 chunks; total ASE vector index 1,726 chunks across Ch.2–3; local OCR on 50/50 scanned spreads).
- [x] Author packs, lint strictly (`note_linter.py --strict` = 0), export DirectWrite PDFs, and audit cohesion (0 split questions).